HAGGAÏ BONNEAU

Paris, France
Google Scholar: https://scholar.google.com/citations?user=iTKDGOEAAAAJ

RESEARCH INTERESTS

Non-equilibrium statistical physics · Stochastic processes · Interacting many-body systems · Electrolytes · Polymer physics · Random graphs and complex networks

RESEARCH EXPERIENCE

Career Break for Family Caregiving 2025–2026

Postdoctoral Researcher — Tel Aviv University & IPhT Saclay

2024–2025

- Developed stochastic field-theory descriptions of concentrated electrolytes and derived analytical predictions for their frequency-dependent conductivity and correlation functions.
- Studied the effects of electrostatic, hard-core, and hydrodynamic interactions on the dynamical response of interacting ionic systems.
- Developed Python implementations for coupled stochastic and deterministic differential equations and used numerical calculations to test analytical approximations.

Ph.D. Researcher — ESPCI Paris, Gulliver Laboratory

2020–2023

- Studied the transient response of driven electrolytes following sudden changes in an external electric field using stochastic density functional theory and fluctuating hydrodynamics.
- Showed how algebraically slow relaxation of the electric current is related to the diffusive dynamics and long-range structure of non-equilibrium correlations.
- Investigated the glass transition of confined polymer films using exact first-passage and path statistics of finite random walks, allowing a proposed relaxation mechanism to be tested against experimental observations in collaboration with the Forrest group at the University of Waterloo.
- Analyzed discrete-element simulations of dense granular flows and connected microscopic particle interactions to macroscopic constitutive behavior.

Research Assistant — Hebrew University of Jerusalem, Racah Institute

2016–2018

- Studied structural properties of large random-graph ensembles using generating-function and message-passing methods.
- Derived analytical results for quantities including shortest-cycle statistics and articulation points.
- Generated and analyzed random-graph ensembles numerically and compared their structural statistics with analytical predictions.

EDUCATION

Ph.D. in Physics — ESPCI Paris, 2023

Gulliver Laboratory
Advisors: Élie Raphaël and Vincent Démery
Thesis: *Currents, correlations and effects of confinement in out-of-equilibrium systems*

M.Sc. in Physics — Hebrew University of Jerusalem, 2020

Graduated with honors · Academic Excellence Award

B.Sc. in Physics — Hebrew University of Jerusalem, 2018

Graduated with honors

TEACHING EXPERIENCE

Teaching & Laboratory Instructor — ESPCI Paris & Hebrew University of Jerusalem

2018–2021

- Teaching assistant for event-driven molecular dynamics simulations at ESPCI Paris.
- Teaching assistant for electromagnetism at the Hebrew University of Jerusalem.
- Laboratory instructor for superconductivity experiments at the Hebrew University of Jerusalem.

ADDITIONAL INFORMATION

Programming: Python (NumPy, SciPy, pandas, SymPy, Matplotlib, familiarity with PyTorch), Mathematica, Linux, HPC clusters
Languages: Hebrew (native), English (fluent), French (fluent), Arabic (basic)

PUBLICATIONS

H. Bonneau, V. Démery, E. Raphaël.
"Stationary and transient correlations in driven electrolytes."
Journal of Statistical Mechanics: Theory and Experiment, 033201 (2025).

H. Bonneau, Y. Avni, D. Andelman, H. Orland.
"Frequency-dependent conductivity of concentrated electrolytes: A stochastic density functional theory."
Journal of Chemical Physics **161**, 244501 (2024).

H. Bonneau, V. Démery, E. Raphaël.
"Temporal response of the conductivity of electrolytes."
Journal of Statistical Mechanics: Theory and Experiment, 073205 (2023).

H. Bonneau, M. Arutkin, R. Chen, J. A. Forrest, E. Raphaël, T. Salez.
"On the bridge hypothesis in the glass transition of freestanding polymer films."
European Physical Journal E **46**, 8 (2023).

D. Dumont, **H. Bonneau**, T. Salez, E. Raphaël, P. Damman.
"Microscopic foundation of the μ(I) rheology for dense granular flows on inclined planes."
Physical Review Research **5**, 013089 (2023).

H. Bonneau, I. Tishby, O. Biham, E. Katzav, R. Kühn.
"Fate of articulation points and bredges in percolation."
Physical Review E **103**, 042302 (2021).

H. Bonneau, O. Biham, R. Kühn, E. Katzav.
"Statistical analysis of edges and bredges in configuration model networks."
Physical Review E **102**, 012314 (2020).

H. Bonneau, A. Hassid, O. Biham, R. Kühn, E. Katzav.
"Distribution of shortest cycle lengths in random networks."
Physical Review E **96**, 062307 (2017).